

AI Voice Agent — Deployment Status

AWS hosting + Exotel telephony integration — progress update, July 9, 2026

Summary

The backend is deployed on AWS and running end-to-end: STT (Sarvam), LLM (Gemini), and TTS (Sarvam) are all live and verified with real API keys, the knowledge base is loaded, and a browser-based test call is working. Source code is in a private GitHub repository. The Exotel telephony gateway is already built into the codebase; the remaining work is infrastructure hardening (static IP, domain, TLS) and Exotel-side configuration to connect real phone calls.

What's Live Right Now

Item	Status
Backend application	FastAPI (Python 3.12), running on AWS EC2 (Ubuntu 24.04 LTS)
Process management	systemd service — auto-restarts on crash or server reboot
Source control	Private GitHub repo: github.com/rounittxx/ai-voice-agent
Health check	Verified: /health returns OK, STT + LLM keys confirmed valid
Knowledge base	Loaded — 33 articles indexed (in-memory retrieval store)
Voice pipeline test	Browser-based test call in progress (mic, STT, LLM, TTS round trip)
Telephony code	Exotel bidirectional WebSocket gateway already implemented; not yet connected to Exotel platform — pending HTTPS setup below

Architecture Basis

All AI processing (speech-to-text, language model, text-to-speech, embeddings) runs through hosted APIs — Sarvam and Google Gemini. The server only handles audio streaming and lightweight signal processing (voice activity detection, noise suppression). **No GPU is required anywhere in this stack** — confirmed against the project's dependencies.

Remaining Steps to Go-Live

- 1. Allocate a static (Elastic) IP — prevents the server address from changing on restart, which would break DNS.
- 2. Add a DNS record on the company domain pointing a subdomain (e.g. [voice.company.com](#)) at the server.
- 3. Open HTTPS (443) on the server's firewall to the public internet.
- 4. Install automatic TLS (Caddy / Let's Encrypt) so the endpoint serves valid HTTPS — required by Exotel, which only accepts secure (wss://) connections.
- 5. Configure the Voicebot Applet in Exotel's App Bazaar to point at the new secure endpoint.
- 6. Run a live test call through an Exotel phone number end-to-end.

- 7. Load-test at 20 concurrent calls, then scale infrastructure (2+ servers behind a load balancer) before full rollout.

Security & Follow-Up Items

- SSH access is currently open broadly for setup convenience — should be restricted to specific IPs, or moved to AWS Session Manager, before this goes near production.
- A temporary GitHub access token was used to push the initial code and should be rotated now that setup is complete.
- Currently a single server with no redundancy — appropriate for this testing phase; production sizing for 20 concurrent calls calls for two servers behind a load balancer (previously scoped separately).
- Data is currently stored in SQLite with mock/seed records — fine for the pilot, should move to a managed database (RDS Postgres) before handling real customer data.

Prepared from the deployment session on July 9, 2026 — covers work completed and the sequence of remaining steps to connect live Exotel calls.